

Project 10 — Disease-Associated Mutations & Protein Structure

Connecting genetic variation with protein structure and function

Research Question

How can a disease-associated mutation be investigated at the protein level to understand its possible structural and functional consequences?

Project Overview

Genetic mutations can alter protein sequences and potentially affect protein stability, molecular interactions, folding, or function. This project introduces a structure-based approach to investigating variants while keeping computational interpretation separate from experimental proof.

Computational Workflow

```
graph TD; A[Disease-associated variant] --> B[Variant / protein identification]; B --> C[Reference protein sequence]; C --> D[Protein structure retrieval]; D --> E[Mutation mapping]; E --> F[3D structural inspection]; F --> G[Functional interpretation]; G --> H[Limitations and hypotheses];
```

Structural Interpretation

A mutation may occur in an α -helix, β -sheet, loop, active-site region, ligand-binding region, protein-protein interface, buried core, or solvent-exposed surface. These observations can help prioritize biological questions but do not by themselves establish pathogenicity.

Sequence-to-Structure Connection

Useful questions include whether the substitution changes charge, size, polarity, hydrophobicity, or local interactions, and whether the residue is conserved among homologous proteins.

Methodological Note

This portfolio version deliberately does not invent a mutation-specific experimental result. A stronger independently reproducible extension would select a documented disease-associated variant, record its accession and residue change, map it to a validated structure, and compare the local structural environment.

Limitations

Static structures represent limited conformational states. Structural location alone cannot prove pathogenicity. Computational predictions require validation, and variant effects may depend on interactions not represented by an isolated structure.

What I Learned

How genetic variation can be connected to protein structure; why amino-acid properties matter; how structural context supports biological reasoning; and why computational observations should be separated from

experimentally demonstrated effects.

Conclusion

Disease-associated mutation analysis provides a bridge between genomics and structural biology. Mapping a variant onto a protein structure can turn a sequence-level observation into a more specific molecular question while keeping claims proportional to available evidence.